

CHARLTON SMITH

M.S. Software Engineering

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SUMMARY

Full-stack engineer and builder with 7+ years shipping production systems, AI-powered applications, and entrepreneurial ventures. From co-founding a computer vision startup to operating a 50-machine NFT trading system to currently building LLM agents and tools for web3 and fintech — I turn ideas into working products. Passionate about AI as a tool to enhance human creativity and developer productivity.

Paste your job description here <https://azoni.ai/chat> and try out my personalized AI chatbot.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, C#

AI/ML: OpenAI APIs (GPT-4), Claude API, LLM Agents, RAG, LangChain, Prompt Engineering

Frontend: React, Vite, HTML, CSS, Mobile-First Responsive Design

Backend: Node.js, FastAPI, Django, REST APIs, Microservices

Cloud: AWS (Lambda, EC2, S3, CloudWatch), Docker, CI/CD, Netlify, Render

Databases: PostgreSQL, MongoDB, Redis, Firebase, SQLite

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Capital One — *Remote*

Nov 2022 – Nov 2023

- Led design of automated testing pipeline validating customer email notifications triggered by credit card activity — 1,000+ test cases stored as JSON in S3, executed via AWS Lambda
- Built Python microservices pulling test configurations from GitHub, running validations against notification outputs, and logging results to CloudWatch for debugging
- Implemented Slack and email integrations for real-time failure alerts, reducing incident response time for engineering teams
- Mentored summer intern from project scoping through code reviews to production deployment; intern shipped internal test automation tool

Software Engineer II | T-Mobile — *Bellevue, WA*

Jun 2018 – Apr 2022

- Designed and built internal automation platform from scratch serving 20-100 users, reducing network operations workload by 80% through workflow automation
- Architected Python/Django backend with modular design; led refactoring effort to microservices architecture as system grew
- Migrated frontend from Django templates to Angular SPA; built reusable component library that accelerated feature delivery across team
- Unified spectrum negotiation documentation onto Hugo-based platform, making critical technical docs maintainable by non-technical stakeholders

Computer Science Instructor | Nucamp Coding Bootcamp — *Seattle, WA*

Jan 2018 – May 2018

- Taught full-stack web development (JavaScript, HTML/CSS, Git, REST APIs) to career-transition students in intensive bootcamp format

Salesforce Engineer | Slalom Consulting — *Seattle, WA*

Nov 2017 – Mar 2018

- Earned three Salesforce certifications (Integration Architect Designer, Administrator, Platform App Builder) and delivered enterprise CRM solutions for Fortune 500 clients

Software Engineering Intern | Givesafe (now Samaritan) — *Seattle, WA*

Dec 2016 – Apr 2017

- Built features in Ruby on Rails for social impact platform connecting donors directly with people experiencing homelessness — now serving 35+ cities nationwide

VENTURES & TECHNICAL PROJECTS

AI & Fintech Applications

Dec 2024 – Present

- Old Ways Today — Full-stack AI chatbot (React, FastAPI, PostgreSQL) helping families find non-toxic products. Integrated OpenAI API with custom system prompts, IP-based rate limiting, token usage logging, and full blog CMS with password-protected admin panel
- DuMarket — Full-stack prediction market webapp with Central Limit Order Book matching engine using price-time priority algorithm, automated market maker with inventory-aware pricing, and real-time P&L tracking. React frontend, FastAPI backend, PostgreSQL. 18 REST endpoints, 6 database models
- LLM-Powered Bots — Discord and Twitter agents with persistent conversation memory (SQLite), tool integration (Google Calendar), and agentic decision-making architectures using GPT-4 and Claude APIs
- azoni.ai — Portfolio site with AI assistant trained on my background via custom system prompts; recruiters can paste job descriptions for automated fit analysis

Rowing Tracker — AI-Powered Fitness App

2024

- Built training app using Claude's multimodal API to extract workout metrics from rowing machine screen photos with human-in-the-loop verification before saving to Firebase
- Implemented gamification features (leaderboards, streaks, achievements) driving engagement among active users; architected for scalability

Dustbunny — NFT Trading Operation

Sep 2021 – Jun 2022

- Built and operated automated NFT bidding system distributed across 50 machines to work around OpenSea rate limits, processing 2,500+ requests per minute across local network
- Engineered real-time floor price polling aggregated to Redis for sub-second read performance; built competitive bidding algorithms with max bid safeguards and liquidity analysis
- Integrated OpenSea SDK for order execution, Etherscan API for blockchain data, and deployed ERC-721 smart contract via Remix

OLI Fitness — Co-Founder & Technical Lead

Jan 2016 – Mar 2018

- Co-founded computer vision fitness startup; built real-time joint tracking system using Microsoft Kinect C# SDK to analyze squat and deadlift form against professional overlays
- Developed movement scoring algorithms comparing user joint angles to "good form" parameters defined with fitness expert consultation
- Published extended abstract at ACM CHI 2017; Princeton Tiger Launch regional finalist; UW Business Plan Competition finalist; selected for Collision Alpha startup program

EDUCATION

M.S. Software Engineering — Colorado Technical University

2021

B.S. Computer Science — University of Washington Tacoma

2017

Graduated with Honors • Teaching Assistant (Java, Python)

RECOGNITION & COMMUNITY

- 1st Place — T-Mobile Big Data Hackathon: Built HashMaps, auto-hashtag tool using OpenCV image analysis and geolocation
- Head Organizer — Global AI Hackathon Seattle 2017: Coordinated sponsors, judges, venue, and 100+ participants
- President — Huscii Coding Club, UW Tacoma: Led weekly coding workshops, organized hackathon teams, ran interview prep sessions; grew active membership and encouraged regular hackathon attendance
- Mentor — Expedia Coding for Kids: Weekly volunteer teaching elementary students programming fundamentals through interactive exercises